

JEE Mains 2019 Chapter wise Question Bank

Matrices and Determinants – Questions

Q1

The system of linear equations

$$x + y + z = 2$$

$$2x + 3y + 2z = 5$$

$$2x + 3y + (a^2 - 1)z = a + 1$$

(1) is inconsistent when $a = 4$ (2) has a unique solution for $|a| = \sqrt{3}$ (3) has infinitely many solutions for $a = 4$ (4) is inconsistent when $|a| = \sqrt{3}$

9 Jan Morning

Q2

If $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, then the matrix A^{-50} when $\theta = \frac{\pi}{12}$,

is equal to:

$$(1) \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \quad (2) \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$(3) \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \quad (4) \begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$$

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Q3

If the system of linear equations

$$x - 4y + 7z = g$$

$$3y - 5z = h$$

$$-2x + 5y - 9z = k$$

is consistent, then :

$$(1) g + 2h + k = 0$$

$$(2) g + h + 2k = 0$$

$$(3) 2g + h + k = 0$$

$$(4) g + h + k = 0$$

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Q4

If

$$A = \begin{bmatrix} e^t & e^{-t} \cos t & e^{-t} \sin t \\ e^t & -e^{-t} \cos t - e^{-t} \sin t & -e^{-t} \sin t + e^{-t} \cos t \\ e^t & 2e^{-t} \sin t & -2e^{-t} \cos t \end{bmatrix}$$

then A is:

(1) invertible for all $t \in \mathbf{R}$.(2) invertible only if $t = \pi$.(3) not invertible for any $t \in \mathbf{R}$.(4) invertible only if $t = \frac{\pi}{2}$.

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Q5

If the system of equations

$$x + y + z = 5$$

$$x + 2y + 3z = 9$$

$$x + 3y + \alpha z = \beta$$

has infinitely many solutions, then $\beta - \alpha$ equals:

(1) 21 (2) 8 (3) 18 (4) 5

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Q6

The number of values of $\theta \in (0, \pi)$ for which the system of linear equations

$$x + 3y + 7z = 0$$

$$-x + 4y + 7z = 0$$

$$(\sin 3\theta)x + (\cos 2\theta)y + 2z = 0$$

has a non-trivial solution, is:

- (1) three (2) two
(3) four (4) one

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Q7

Let $A = \begin{bmatrix} 2 & b & 1 \\ b & b^2 + 1 & b \\ 1 & b & 2 \end{bmatrix}$ where $b > 0$. Then the minimum

value of $\frac{\det(A)}{b}$ is:

- (1) $2\sqrt{3}$ (2) $-2\sqrt{3}$
(3) $-\sqrt{3}$ (4) $\sqrt{3}$

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Q8

If the system of linear equations

$$2x + 2y + 3z = a$$

$$3x - y + 5z = b$$

$$x - 3y + 2z = c$$

where, a, b, c are non-zero real numbers, has more than one solution, then :

- (1) $b - c + a = 0$ (2) $b - c - a = 0$
(3) $a + b + c = 0$ (4) $b + c - a = 0$

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Q9

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Let $A = \begin{pmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{pmatrix}$. If $AA^T = I_3$, then $|p|$ is :

- (1) $\frac{1}{\sqrt{5}}$ (2) $\frac{1}{\sqrt{3}}$
(3) $\frac{1}{\sqrt{2}}$ (4) $\frac{1}{\sqrt{6}}$

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Q10

If $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$

$= (a+b+c)(x+a+b+c)^2$, $x \neq 0$ and $a+b+c \neq 0$, then x is equal to :

- (1) abc (2) $-(a+b+c)$
(3) $2(a+b+c)$ (4) $-2(a+b+c)$

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Q11

Let A and B be two invertible matrices of order 3×3 . If $\det(ABA^T) = 8$ and $\det(AB^{-1}) = 8$, then $\det(BA^{-1}B^T)$ is equal to :

- (1) $\frac{1}{4}$ (2) 1
(3) $\frac{1}{16}$ (4) 16

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Q12

An ordered pair (α, β) for which the system of linear equations

$$(1 + \alpha)x + \beta y + z = 2$$

$$ax + (1 + \beta)y + z = 3$$

$$\alpha x + \beta y + 2z = 2$$

has a unique solution, is :

- (1) (2, 4) (2) (-3, 1)
(3) (-4, 2) (4) (1, -3)

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Q13

Matrices and Determinants

Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $Q = [q_{ij}]$ be two 3×3 matrices such

that $Q - P^5 = I_3$. Then $\frac{q_{21} + q_{31}}{q_{32}}$ is equal to :

- (1) 10 (2) 135
(3) 15 (4) 9

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Q14

If $A = \begin{bmatrix} 1 & \sin\theta & 1 \\ -\sin\theta & 1 & \sin\theta \\ -1 & -\sin\theta & 1 \end{bmatrix}$; then for all $\theta \in \left(\frac{3\pi}{4}, \frac{5\pi}{4}\right)$,

$\det(A)$ lies in the interval :

- (1) $\left(1, \frac{5}{2}\right]$ (2) $\left[\frac{5}{2}, 4\right)$
(3) $\left(0, \frac{3}{2}\right]$ (4) $\left(\frac{3}{2}, 3\right]$

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Q15

The set of all values of λ for which the system of linear equations

$$x - 2y - 2z = \lambda x$$

$$x + 2y + z = \lambda y$$

$$-x - y = \lambda z$$

has a non-trivial solution :

- (1) is a singleton
(2) contains exactly two elements
(3) is an empty set
(4) contains more than two elements

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Q16

The greatest value of $c \in \mathbb{R}$ for which the system of linear equations

$$x - cy - cz = 0$$

$$cx - y + cz = 0$$

$$cx + cy - z = 0$$

has a non-trivial solution, is :

- (1) -1 (2) $\frac{1}{2}$ (3) 2 (4) 0

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Q17

Let $A = \begin{bmatrix} \cos\alpha - \sin\alpha & \\ \sin\alpha & \cos\alpha \end{bmatrix}$, ($\alpha \in \mathbb{R}$) such that $A^{32} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$.

Then a value of α is :

- (1) $\frac{\pi}{32}$ (2) 0 (3) $\frac{\pi}{64}$ (4) $\frac{\pi}{16}$

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Q18

Let the numbers 2, b , c be in an A.P. and

$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & b & c \\ 4 & b^2 & c^2 \end{bmatrix}$. If $\det(A) \in [2, 16]$, then c lies in the

interval :

- (1) $[2, 3]$ (2) $(2 + 2^{3/4}, 4)$
(3) $[4, 6]$ (4) $[3, 2 + 2^{3/4}]$

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Q19

If the system of linear equations

$$x - 2y + kz = 1$$

$$2x + y + z = 2$$

$$3x - y - kz = 3$$

has a solution (x, y, z) , $z \neq 0$, then (x, y) lies on the straight line whose equation is :

- (1) $3x - 4y - 1 = 0$ (2) $4x - 3y - 4 = 0$
(3) $4x - 3y - 1 = 0$ (4) $3x - 4y - 4 = 0$

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Q20

If

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix},$$

then the inverse of $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is:

- (1) $\begin{bmatrix} 1 & 0 \\ 12 & 1 \end{bmatrix}$ (2) $\begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$
(3) $\begin{bmatrix} 1 & -12 \\ 0 & 1 \end{bmatrix}$ (4) $\begin{bmatrix} 1 & 0 \\ 13 & 1 \end{bmatrix}$

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Q21

Let α and β be the roots of the equation $x^2 + x + 1 = 0$. Then for $y \neq 0$ in $\mathbb{R}$,

$$\begin{vmatrix} y+1 & \alpha & \beta \\ \alpha & y+\beta & 1 \\ \beta & 1 & y+\alpha \end{vmatrix} \text{ is equal to:}$$

- (1) $y(y^2 - 1)$ (2) $y(y^2 - 3)$
 (3) y^3 (4) $y^3 - 1$

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Q22

The total number of matrices $A = \begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}$,

($x, y \in \mathbb{R}, x \neq y$) for which $A^T A = 3I_3$ is:

- (1) 2 (2) 3 (3) 6 (4) 4

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Q23

If the system of equations $2x + 3y - z = 0$, $x + ky - 2z = 0$ and $2x - y + z = 0$ has a non-trivial solution (x, y, z), then

$\frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k$ is equal to:

- (1) $\frac{3}{4}$ (2) $\frac{1}{2}$ (3) $-\frac{1}{4}$ (4) -4

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Q24

If $\Delta_1 = \begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$ and

$$\Delta_2 = \begin{vmatrix} x & \sin 2\theta & \cos 2\theta \\ -\sin 2\theta & -x & 1 \\ \cos 2\theta & 1 & x \end{vmatrix}, x \neq 0;$$

then for all $\theta \in \left(0, \frac{\pi}{2}\right)$:

- (1) $\Delta_1 - \Delta_2 = -2x^3$
 (2) $\Delta_1 - \Delta_2 = x(\cos 2\theta - \cos 4\theta)$
 (3) $\Delta_1 \times \Delta_2 = -2(x^3 + x - 1)$
 (4) $\Delta_1 + \Delta_2 = -2x^3$

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Q25

If the system of linear equations

$$x + y + z = 5$$

$$x + 2y + 2z = 6$$

$x + 3y + \lambda z = \mu$, ($\lambda, \mu \in \mathbb{R}$), has infinitely many solutions, then the value of $\lambda + \mu$ is :

- (1) 12 (2) 9 (3) 7 (4) 10

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Q26

Let λ be a real number for which the system of linear equations:

$$x + y + z = 6$$

$$4x + \lambda y - \lambda z = \lambda - 2$$

$$3x + 2y - 4z = -5$$

has infinitely many solutions. Then λ is a root of the quadratic equation :

- (1) $\lambda^2 + 3\lambda - 4 = 0$ (2) $\lambda^2 - 3\lambda - 4 = 0$
 (3) $\lambda^2 + \lambda - 6 = 0$ (4) $\lambda^2 - \lambda - 6 = 0$

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Q27

The sum of the real roots of the equation

$$\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0, \text{ is equal to :}$$

- (1) 6 (2) 0 (3) 1 (4) -4

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Q28

If A is a symmetric matrix and B is a skew-symmetric matrix such that $A + B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$, then AB is equal to :

- (1) $\begin{bmatrix} -4 & -1 \\ -1 & 4 \end{bmatrix}$ (2) $\begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$
 (3) $\begin{bmatrix} 4 & -2 \\ 1 & -4 \end{bmatrix}$ (4) $\begin{bmatrix} -4 & 2 \\ 1 & 4 \end{bmatrix}$

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Q29

If $B = \begin{bmatrix} 5 & 2\alpha & 1 \\ 0 & 2 & 1 \\ \alpha & 3 & -1 \end{bmatrix}$ is the inverse of a 3×3 matrix A, then

the sum of all values of α for which $\det(A) + 1 = 0$, is :

- (1) 0 (2) -1 (3) 1 (4) 2

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Matrices and Determinants - Solutions

Q1

(4) Since the system of linear equations are

$$x + y + z = 2 \quad \dots(1)$$

$$2x + 3y + 2z = 5 \quad \dots(2)$$

$$2x + 3y + (a^2 - 1)z = a + 1 \quad \dots(3)$$

Now, $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 2 & 3 & a^2 - 1 \end{vmatrix}$

$$\Rightarrow \Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 0 & 0 & a^2 - 3 \end{vmatrix}$$

(Applying $R_3 \rightarrow R_3 - R_2$)

$$= a^2 - 3$$

$$\text{When, } \Delta = 0 \Rightarrow a^2 - 3 = 0 \Rightarrow |a| = \sqrt{3}$$

If $a^2 = 3$, then plane represented by eqn (2) and eqn (3) are parallel.

Hence, the given system of equation is inconsistent.

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Q2

$$(3) \quad A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \Rightarrow |A| = 1$$

$$\text{adj}(A) = \begin{bmatrix} +\cos \theta & -\sin \theta \\ +\sin \theta & +\cos \theta \end{bmatrix}^T = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = B$$

$$B^2 = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{bmatrix}$$

$$\Rightarrow B^3 = \begin{bmatrix} \cos 3\theta & \sin 3\theta \\ -\sin 3\theta & \cos 3\theta \end{bmatrix}$$

$$\Rightarrow A^{-50} = B^{50} = \begin{bmatrix} \cos(50\theta) & \sin(50\theta) \\ -\sin(50\theta) & \cos(50\theta) \end{bmatrix}$$

$$(A^{-50})_{\theta=\frac{\pi}{12}} = \begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$\left[\because \cos\left(\frac{50\pi}{12}\right) = \cos\left(4\pi + \frac{\pi}{6}\right) = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2} \right]$$

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Q3

(3) Consider the system of linear equations

$$x - 4y + 7z = g \quad \dots(i)$$

$$3y - 5z = h \quad \dots(ii)$$

$$-2x + 5y - 9z = k \quad \dots(iii)$$

Multiply equation (i) by 2 and add equation (i), equation (ii) and equation (iii)

$$\Rightarrow 0 = 2g + h + k. \therefore 2g + h + k = 0$$

then system of equation is consistent.

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Q4

$$(1) \det(A) = |A|$$

$$= \begin{vmatrix} e^t & e^{-t} \cos t & e^{-t} \sin t \\ e^t & -e^{-t} \cos t - e^{-t} \sin t & -e^{-t} \sin t + e^{-t} \cos t \\ e^t & 2e^{-t} \sin t & -2e^{-t} \cos t \end{vmatrix}$$

$$= e^t \cdot e^{-t} \cdot e^{-t} \begin{vmatrix} 1 & \cos t & \sin t \\ 1 & -\cos t - \sin t & -\sin t + \cos t \\ 1 & 2 \sin t & -2 \cos t \end{vmatrix}$$

$$= e^{-t} \begin{vmatrix} 0 & 2 \cos t + \sin t & 2 \sin t - \cos t \\ 0 & -\cos t - 3 \sin t & -\sin t + 3 \cos t \\ 1 & 2 \sin t & -2 \cos t \end{vmatrix} \begin{matrix} R_1 \rightarrow R_1 - R_2 \\ R_2 \rightarrow R_2 + R_3 \end{matrix}$$

$$= e^{-t} \begin{vmatrix} 0 & -5 \sin t & 5 \cos t \\ 0 & -\cos t - 3 \sin t & -\sin t + 3 \cos t \\ 1 & 2 \sin t & -2 \cos t \end{vmatrix} \begin{matrix} R_1 \rightarrow R_1 + 2R_2 \end{matrix}$$

$$= e^{-t} [(-5 \sin t)(-\sin t + 3 \cos t) - 5 \cos t (-\cos t - 3 \sin t)]$$

$$= 5e^{-t} \neq 0, \forall t \in R$$

$\therefore A$ is invertible.

$$\int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx, x \geq 0$$

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Q5

$$(2) \Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & \alpha \end{vmatrix} = 2\alpha - 9 - \alpha + 3 + 1 = \alpha - 5$$

$$\Delta_1 = \begin{vmatrix} 5 & 1 & 1 \\ 9 & 2 & 3 \\ \beta & 3 & \alpha \end{vmatrix}$$

$$= 5(2\alpha - 9) - 1(9\alpha - 3\beta) + (27 - 2\beta)$$

$$= 10\alpha - 45 - 9\alpha + 3\beta + 27 - 2\beta$$

$$= \alpha + \beta - 18$$

$$\Delta_2 = \begin{vmatrix} 1 & 5 & 1 \\ 1 & 9 & 3 \\ 1 & \beta & \alpha \end{vmatrix}$$

$$= 9\alpha - 3\beta - 5(\alpha - 3) + 1(\beta - 9)$$

$$= 9\alpha - 3\beta - 5\alpha + 15 + \beta - 9 = 4\alpha - 2\beta + 6$$

$$\Delta_3 = \begin{vmatrix} 1 & 1 & 5 \\ 1 & 2 & 9 \\ 1 & 3 & \beta \end{vmatrix} = 2\beta - 27 - \beta + 9 + 5 = \beta - 13$$

Since, the system of equations has infinite many solutions. Hence,

$$\Delta_1 = \Delta_2 = \Delta_3 = \Delta = 0$$

$$\Rightarrow \alpha = 5, \beta = 13 \Rightarrow \beta - \alpha = 8$$

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Q6

Matrices and Determinants

(2) Since, the system of linear equations has, non-trivial solution then determinant of coefficient matrix = 0

$$\text{i.e., } \begin{vmatrix} \sin 3\theta & \cos 2\theta & 2 \\ 1 & 3 & 7 \\ -1 & 4 & 7 \end{vmatrix} = 0$$

$$\sin 3\theta(21 - 28) - \cos 2\theta(7 + 7) + 2(4 + 3) = 0$$

$$\sin 3\theta + 2\cos 2\theta - 2 = 0$$

$$3\sin\theta - 4\sin^3\theta + 2 - 4\sin^2\theta - 2 = 0$$

$$4\sin^3\theta + 4\sin^2\theta - 3\sin\theta = 0$$

$$\sin\theta(4\sin^2\theta + 4\sin\theta - 3) = 0$$

$$\sin\theta(4\sin^2\theta + 6\sin\theta - 2\sin\theta - 3) = 0$$

$$\sin\theta[2\sin\theta(2\sin\theta - 1) + 3(2\sin\theta - 1)] = 0$$

$$\sin\theta(2\sin\theta - 1)(2\sin\theta + 3) = 0$$

$$\sin\theta = 0, \sin\theta = \frac{1}{2}$$

$$\theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\left(\because \sin\theta \neq -\frac{3}{2} \right)$$

Hence, for two values of θ , system of equations has non-trivial solution

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Q7

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$$\begin{aligned} (1) |A| &= \begin{vmatrix} 2 & b & 1 \\ b & b^2 + 1 & b \\ 1 & b & 2 \end{vmatrix} \\ &= 2(2b^2 + 2 - b^2) - b(2b - b) + 1(b^2 - b^2 - 1) \\ &= 2b^2 + 4 - b^2 - 1 = b^2 + 3 \end{aligned}$$

$$\frac{|A|}{b} = b + \frac{3}{b}$$

$$\therefore \frac{b + \frac{3}{b}}{2} \geq \left(b \frac{3}{b}\right)^{\frac{1}{2}} \Rightarrow b + \frac{3}{b} \geq 2\sqrt{3}$$

$$\therefore \frac{|A|}{b} \geq 2\sqrt{3}$$

Minimum value of $\frac{|A|}{b}$ is $2\sqrt{3}$.

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Q8

(2) $\therefore$ System of equations has more than one solution

$\therefore \Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$ for infinite solution

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} a & 2 & 3 \\ b & -1 & 5 \\ c & -3 & 2 \end{vmatrix} = a(13) + 2(5c - 2b) + 3(-3b + c) \\ &= 13a - 13b + 13c = 0 \end{aligned}$$

$$\text{i.e., } a - b + c = 0$$

$$\text{or } b - c - a = 0$$

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Q9

Matrices and Determinants

$$(3) A = \begin{bmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{bmatrix}$$

$$\begin{aligned} \therefore A \cdot A^T &= \begin{bmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{bmatrix} \times \begin{bmatrix} 0 & p & p \\ 2q & q & -q \\ r & -r & r \end{bmatrix} \\ &= \begin{bmatrix} 4q^2 + r^2 & 2q^2 - r^2 & -2q^2 + r^2 \\ 2q^2 - r^2 & p^2 + q^2 + r^2 & p^2 - q^2 - r^2 \\ -2q^2 + r^2 & p^2 - q^2 - r^2 & p^2 + q^2 + r^2 \end{bmatrix} \end{aligned}$$

Given, $AA^T = I$

$$\therefore 4q^2 + r^2 = p^2 + q^2 + r^2 = 1$$

$$\Rightarrow p^2 - 3q^2 = 0 \text{ and } r^2 = 1 - 4q^2$$

$$\text{and } 2q^2 - r^2 = 0 \Rightarrow r^2 = 2q^2$$

$$\therefore p^2 = \frac{1}{2}, q^2 = \frac{1}{6} \text{ and } r^2 = \frac{1}{3}$$

$$\therefore |p| = \frac{1}{\sqrt{2}}$$

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Q10

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$$(4) \Delta = \begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_2 + R_3$$

$$\Delta = \begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$= (a+b+c) \begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$C_1 \rightarrow C_1 - C_3, C_2 \rightarrow C_2 - C_3$$

$$\Delta = (a+b+c) \begin{vmatrix} 0 & 0 & 1 \\ 0 & -b-c-a & 2b \\ c+a+b & c+a+b & c-a-b \end{vmatrix}$$

$$= (a+b+c) (a+b+c)^2$$

$$\text{Hence, } x = -2(a+b+c)$$

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Q11

$$(3) \text{ Let } |A| = a, |B| = b$$

$$\Rightarrow |A^T| = a |A^{-1}| = \frac{1}{a}, |B^T| = b, |B^{-1}| = \frac{1}{b}$$

$$\therefore |ABA^T| = 8 \Rightarrow |A| |B| |A^T| = 8 \quad \dots(1)$$

$$\Rightarrow a \cdot b \cdot a = 8 \Rightarrow a^2 b = 8$$

$$\therefore |AB^{-1}| = 8 \Rightarrow |A| |B^{-1}| = 8 \Rightarrow a \cdot \frac{1}{b} = 8 \quad \dots(2)$$

From (1) & (2)

$$a = 4, b = \frac{1}{2}$$

$$\text{Then, } |BA^{-1}B^T| = |B| |A^{-1}| |B^T| = b \cdot \frac{1}{a} \cdot b = \frac{b^2}{a} = \frac{1}{16}$$

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Q12

Matrices and Determinants

(1) $\therefore$ The system of linear equations has a unique solution.

$$\therefore \Delta \neq 0$$

$$\Delta = \begin{vmatrix} 1+\alpha & \beta & 1 \\ \alpha & 1+\beta & 1 \\ \alpha & \beta & 2 \end{vmatrix} \neq 0$$

$$\Rightarrow \begin{vmatrix} 1+\alpha+\beta+1 & \beta & 1 \\ \alpha+1+\beta+1 & \beta+1 & 1 \\ \alpha+\beta+2 & \beta & 2 \end{vmatrix} \neq 0 \quad [C_1 \rightarrow C_1 + C_2 + C_3]$$

$$\Rightarrow (\alpha+\beta+2) \begin{vmatrix} 1 & \beta & 1 \\ 1 & \beta+1 & 1 \\ 1 & \beta & 2 \end{vmatrix} \neq 0$$

$$\Rightarrow (\alpha+\beta+2) \begin{vmatrix} 1 & \beta & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \neq 0 \quad \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\Rightarrow (\alpha+\beta+2) 1(1) \neq 0$$

$$\Rightarrow \alpha+\beta+2 \neq 0$$

$\therefore$ Ordered pair (2, 4) satisfies this condition

$$\therefore \alpha = 2 \text{ and } \beta = 4.$$

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Q13

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$$(1) P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 27 & 6 & 1 \end{bmatrix}$$

$$\Rightarrow P^4 = \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 27 & 6 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 6 & 1 & 0 \\ 27 & 6 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ 90 & 12 & 1 \end{bmatrix}$$

$$\Rightarrow P^5 = \begin{bmatrix} 1 & 0 & 0 \\ 12 & 1 & 0 \\ 90 & 12 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 15 & 1 & 0 \\ 135 & 15 & 1 \end{bmatrix}$$

$$\therefore Q - P^5 = I_3$$

$$\therefore Q = I_3 + P^5 = \begin{bmatrix} 2 & 0 & 0 \\ 15 & 2 & 0 \\ 135 & 15 & 2 \end{bmatrix}$$

$$\frac{q_{21} + q_{31}}{q_{32}} = \frac{15 + 135}{15} = 10$$

12 Jan Morning

Q14

$$(4) |A| = \begin{vmatrix} 1 & \sin \theta & 1 \\ -\sin \theta & 1 & \sin \theta \\ -1 & -\sin \theta & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & 0 & 2 \\ -\sin \theta & 1 & \sin \theta \\ -1 & -\sin \theta & 1 \end{vmatrix} \quad R_1 \rightarrow R_1 + R_3$$

$$= 2(\sin^2 \theta + 1)$$

$$\text{Since, } \theta \in \left(\frac{3\pi}{4}, \frac{5\pi}{4} \right) \Rightarrow \sin^2 \theta \in \left(0, \frac{1}{2} \right)$$

$$\therefore \det(A) \in [2, 3)$$

$$[2, 3) \subset \left(\frac{3}{2}, 3 \right]$$

12 Jan Evening

Q15

Matrices and Determinants

(1) Consider the given system of linear equations

$$x(1-\lambda) - 2y - 2z = 0$$

$$x + (2-\lambda)y + z = 0$$

$$-x - y - \lambda z = 0$$

Now, for a non-trivial solution, the determinant of coefficient matrix is zero.

$$\begin{vmatrix} 1-\lambda & -2 & -2 \\ 1 & 2-\lambda & 1 \\ -1 & -1 & -\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda)^3 = 0$$

$$\lambda = 1$$

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Q16

(2) If the system of equations has non-trivial solutions, then the determinant of coefficient matrix is zero

$$\begin{vmatrix} 1 & -c & -c \\ c & -1 & c \\ c & c & -1 \end{vmatrix} = 0$$

$$\Rightarrow (1-c^2) + c(-c-c^2) - c(c^2+c) = 0$$

$$\Rightarrow (1+c)(1-c) - 2c^2(1+c) = 0$$

$$\Rightarrow (1+c)(1-c-2c^2) = 0$$

$$\Rightarrow (1+c)^2(1-2c) = 0$$

$$\Rightarrow c = -1 \text{ or } \frac{1}{2}$$

Hence, the greatest value of c is $\frac{1}{2}$ for which the system of linear equations has non-trivial solution.

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Q17

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$$(3) A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

$$A^2 = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$$

$$= \begin{bmatrix} \cos 2\alpha & -\sin 2\alpha \\ \sin 2\alpha & \cos 2\alpha \end{bmatrix}$$

$$\text{Similarly, } A^4 = A^2 \cdot A^2 = \begin{bmatrix} \cos 4\alpha & -\sin 4\alpha \\ \sin 4\alpha & \cos 4\alpha \end{bmatrix}$$

$$\text{and so on } A^{32} = \begin{bmatrix} \cos 32\alpha & -\sin 32\alpha \\ \sin 32\alpha & \cos 32\alpha \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

Then $\sin 32\alpha = 1$ and $\cos 32\alpha = 0$

$$\Rightarrow 32\alpha = n\pi + (-1)^n \frac{\pi}{2} \text{ and } 32\alpha = 2n\pi + \frac{\pi}{2}$$

$$\Rightarrow \alpha = \frac{n\pi}{32} + (-1)^n \frac{\pi}{64} \text{ and } \alpha = \frac{n\pi}{16} + \frac{\pi}{64} \text{ where } n \in \mathbb{Z}$$

$$\text{Put } n = 0, \alpha = \frac{\pi}{64}$$

Hence, required value of α is $\frac{\pi}{64}$.

8 April Morning

Q18

$$(3) \text{ Consider, } |A| = \begin{vmatrix} 1 & 1 & 1 \\ 2 & b & c \\ 4 & b^2 & c^2 \end{vmatrix}$$

$$C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_1$$

$$|A| = \begin{vmatrix} 1 & 0 & 0 \\ 2 & b-2 & c-2 \\ 4 & (b-2)(b+2) & (c-2)(c+2) \end{vmatrix}$$

$$= (b-2)(c-2)(c-b)$$

$\therefore 2, b, c$ are in A.P.

$$\therefore (b-2) = (c-b) = d \text{ and } c-2 = 2d$$

$$\Rightarrow |A| = d \cdot 2d \cdot d = 2d^3$$

$$\therefore |A| \in [2, 16] \Rightarrow 1 \leq d^3 \leq 8 \Rightarrow 1 \leq d \leq 2$$

$$4 \leq 2d + 2 \leq 6 \Rightarrow 4 \leq c \leq 6$$

8 April Evening

Q19

Matrices and Determinants

(2) Given system of linear equations,

$$x - 2y + kz = 1, 2x + y + z = 2, 3x - y - kz = 3$$

$$\Delta = \begin{vmatrix} 1 & -2 & k \\ 2 & 1 & 1 \\ 3 & -1 & -k \end{vmatrix}$$

$$= 1(-k+1) + 2(-2k-3) + k(-2-3) \\ = -k+1-4k-6-5k = -10k-5 = -5(2k+1)$$

$$\Delta_1 = \begin{vmatrix} 1 & -2 & k \\ 2 & 1 & 1 \\ 3 & -1 & -k \end{vmatrix} = -5(2k+1)$$

$$\Delta_2 = \begin{vmatrix} 1 & 1 & k \\ 2 & 2 & 1 \\ 3 & 3 & -k \end{vmatrix} = 0, \Delta_3 = \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & 2 \\ 3 & -1 & 3 \end{vmatrix} = 0$$

$$\because z \neq 0 \Rightarrow \Delta = 0$$

$$\Rightarrow -5(2k+1) = 0 \Rightarrow k = -\frac{1}{2}$$

$\therefore$ System of equation has infinite many solutions.

$$\text{Let } z = \lambda \neq 0 \text{ then } x = \frac{10-3\lambda}{10} \text{ and } y = -\frac{2\lambda}{5}$$

$$\therefore (x, y) \text{ must lie on line } 4x - 3y - 4 = 0$$

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Q20

$$(2) \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} \cdots \begin{bmatrix} 1 & n-1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 1+2+3+\dots+(n-1) \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 78 \\ 0 & 1 \end{bmatrix}$$

$$\Rightarrow \frac{(n-1)n}{2} = 78 \Rightarrow n^2 - n - 15 = 0$$

$$\Rightarrow n = 13$$

$$\text{Now, the matrix } \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 13 \\ 0 & 1 \end{bmatrix}$$

$$\text{Then, the required inverse of } \begin{bmatrix} 1 & 13 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -13 \\ 0 & 1 \end{bmatrix}$$

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Q21

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(3) Let $\alpha = \omega$ and $\beta = \omega^2$ are roots of $x^2 + x + 1 = 0$

$$\& \text{ Let } \Delta = \begin{vmatrix} y+1 & \omega & \omega^2 \\ \omega & y+\omega^2 & 1 \\ \omega^2 & 1 & y+\omega \end{vmatrix} = \Delta$$

Applying $C_1 \rightarrow C_1 + C_2 + C_3$, we get

$$\Delta = \begin{vmatrix} y+1+\omega+\omega^2 & \omega & \omega^2 \\ y+1+\omega+\omega^2 & y+\omega^2 & 1 \\ 1+\omega+\omega^2+y & 1 & y+\omega \end{vmatrix}$$

$$\Delta = \begin{vmatrix} y & \omega & \omega^2 \\ y & y+\omega^2 & 1 \\ y & 1 & y+\omega \end{vmatrix} (\because 1+\omega+\omega^2 = 0)$$

$$\Delta = y \begin{vmatrix} 1 & \omega & \omega^2 \\ 1 & y+\omega^2 & 1 \\ 1 & 1 & y+\omega \end{vmatrix}$$

Applying $R_2 \rightarrow R_2 - R_1$ & $R_3 \rightarrow R_3 - R_1$, we get

$$\Delta = y \begin{vmatrix} y+\omega^2-\omega & 1-\omega^2 \\ 1-\omega & y+\omega-\omega^2 \end{vmatrix}$$

$$\Rightarrow \Delta = y[y - (\omega - \omega^2)(y + (\omega - \omega^2)) - (1 - \omega)(1 - \omega^2)]$$

$$\Rightarrow \Delta = y[y^2 - (\omega - \omega^2)^2 - 1 + \omega^2 + \omega - \omega^3]$$

$$\Rightarrow \Delta = y[y^2 - \omega^2 - \omega^4 + 2\omega^3 - 1 + \omega^2 + \omega^4 - \omega^3] (\because \omega^4 = \omega)$$

$$\Rightarrow \Delta = y(y^2) = y^3$$

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Q22

Matrices and Determinants

(4) Given, $A^T A = 3I$

$$\begin{bmatrix} 0 & 2x & 2x \\ 2y & y & -y \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{bmatrix} = 3I$$

$$\Rightarrow \begin{bmatrix} 8x^2 & 0 & 0 \\ 0 & 6y^2 & 0 \\ 0 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\Rightarrow 8x^2 = 3 \text{ and } 6y^2 = 3 \Rightarrow x = \pm\sqrt{\frac{3}{8}} \text{ and } y = \pm\sqrt{\frac{1}{2}}$$

$$\text{Number of combinations of } (x, y) = 2 \times 2 = 4$$

9 April Evening

Q23

(2) Given system of equations has a non-trivial solution.

$$\Rightarrow \Delta = 0 \Rightarrow \begin{vmatrix} 2 & 3 & -1 \\ 1 & k & -2 \\ 2 & -1 & 1 \end{vmatrix} = 0 \Rightarrow k = \frac{9}{2}$$

$$\therefore \text{equations are } \begin{aligned} 2x + 3y - z &= 0 & \dots(i) \\ 2x - y + z &= 0 & \dots(ii) \\ 2x + 9y - 4z &= 0 & \dots(iii) \end{aligned}$$

$$\text{By (i) - (ii), } 2y = z$$

$$\therefore z = -4x \text{ and } 2x + y = 0$$

$$\therefore \frac{x}{y} + \frac{y}{z} + \frac{z}{x} + k = \frac{-1}{2} + \frac{1}{-4} - 4 + \frac{9}{2} = \frac{1}{2}$$

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Q24

$$(4) \Delta_1 = \begin{vmatrix} x & \sin \theta & \cos \theta \\ -\sin \theta & -x & 1 \\ \cos \theta & 1 & x \end{vmatrix}$$

$$= (x - x^2 - 1) - \sin \theta (-x \sin \theta - \cos \theta) + \cos \theta (-\sin \theta + x \cos \theta)$$

$$= -x^3 - x + x \sin^2 \theta + \sin \theta \cos \theta - \cos \theta \sin \theta + x \cos^2 \theta$$

$$= -x^3 - x + x = -x^3$$

$$\text{Similarly, } \Delta_2 = -x^3 \quad \text{Then, } \Delta_1 + \Delta_2 = -2x^3$$

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Q25

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(4) Given system of linear equations: $x + y + z = 5$;
 $x + 2y + 2z = 6$ and $x + 3y + \lambda z = \mu$ have infinite solution.
 $\therefore \Delta = 0, \Delta_x = \Delta_y = \Delta_z = 0$

$$\text{Now, } \Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 3 & \lambda \end{vmatrix} = 0$$

$$\Rightarrow 1(2\lambda - 6) - 1(\lambda - 2) + 1(3 - 2) = 0$$

$$\Rightarrow 2\lambda - 6 - \lambda + 2 + 1 = 0 \Rightarrow \lambda = 3$$

$$\Delta_y = \begin{vmatrix} 1 & 5 & 1 \\ 1 & 6 & 2 \\ 1 & \mu & 3 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} 1 & 5 & 1 \\ 0 & 1 & 1 \\ 0 & \mu - 5 & 2 \end{vmatrix} = 0$$

$$\Rightarrow 1(2 - \mu + 5) = 0 \Rightarrow \mu = 7$$

$$\therefore \lambda + \mu = 10$$

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Q26

(4) $\therefore$ system of equations has infinitely many solutions.
 $\therefore \Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$

$$\text{Here, } \Delta = \begin{vmatrix} 1 & 1 & 1 \\ 4 & \lambda & -\lambda \\ 3 & 2 & -4 \end{vmatrix} = 0$$

$$C_1 \rightarrow C_1 - C_2 \text{ \& } C_2 \rightarrow C_2 - C_3$$

$$\Rightarrow \begin{vmatrix} 0 & 0 & 1 \\ 4 - \lambda & 2\lambda & -\lambda \\ 1 & 6 & -4 \end{vmatrix} = 0 \Rightarrow \lambda = 3$$

$$\text{Now, for } \lambda = 3, \Delta_1 = \begin{vmatrix} 6 & 1 & 1 \\ \lambda - 2 & \lambda & -\lambda \\ -5 & 2 & -4 \end{vmatrix} = 0$$

$$\text{For } \lambda = 3, \Delta_2 = \begin{vmatrix} 1 & 6 & 1 \\ 4 & \lambda - 2 & -\lambda \\ 3 & -5 & -4 \end{vmatrix} = 0$$

$$\text{For } \lambda = 3, \Delta_3 = \begin{vmatrix} 1 & 1 & 6 \\ 4 & \lambda & \lambda - 2 \\ 3 & 2 & -5 \end{vmatrix} = 0$$

$\therefore$ for $\lambda = 3$, system of equations has infinitely many solutions.

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Q27

(2) Given $\begin{vmatrix} x & -6 & -1 \\ 2 & -3x & x-3 \\ -3 & 2x & x+2 \end{vmatrix} = 0$

On expanding,

$$x(-3x^2 - 6x - 2x^2 + 6x) - 6(-3x + 9 - 2x - 4) - (4x - 9x) = 0$$

$$\Rightarrow x(-5x^2) - 6(-5x + 5) - 4x + 9x = 0$$

$$\Rightarrow x^3 - 7x + 6 = 0$$

$\therefore$ all the roots are real.

$$\therefore \text{sum of real roots} = \frac{0}{1} = 0$$

10 April Evening

Q28

(2) Let $A = \begin{bmatrix} a & c \\ c & b \end{bmatrix}$ and $B = \begin{bmatrix} 0 & d \\ -d & 0 \end{bmatrix}$

Then, $A + B = \begin{bmatrix} a & c+d \\ c-d & b \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$

On comparing each term,

$$a = 2, b = -1, c - d = 5, c + d = 3$$

$$\Rightarrow a = 2, b = -1, c = 4, d = -1$$

Now, $AB = \begin{bmatrix} 2 & 4 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$

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Q29

(3) $\therefore B = A^{-1} \Rightarrow |B| = \frac{1}{|A|}$

Now, $|B| = \begin{vmatrix} 5 & 2\alpha & 1 \\ 0 & 2 & 1 \\ \alpha & 3 & -1 \end{vmatrix} = 2\alpha^2 - 2\alpha - 25$

Given, $\det(A) + 1 = 0$

$$\Rightarrow \frac{1}{2\alpha^2 - 2\alpha - 25} + 1 = 0$$

$$\Rightarrow \frac{2\alpha^2 - 2\alpha - 24}{2\alpha^2 - 2\alpha - 25} = 0$$

$$\Rightarrow \alpha = 4, -3 \Rightarrow \text{Sum of values} = 1$$

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